Activity 3.2: Writing Equations of Lines

A

Name: _____ Partner's name: _____ Date: _____

Directions: Write the equation of each line in slope-intercept form, $y = mx + b$.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1

Slope 2, y-intercept -3

Answer: _____ Match? ☐ Yes ☐ No

2

Slope 3, passes through $(1, 5)$

Answer: _____ Match? ☐ Yes ☐ No

3

Passes through $(0, 4)$ and $(2, 10)$

Answer: _____ Match? ☐ Yes ☐ No

4

Passes through $(1, 1)$ and $(3, 5)$

Answer: _____ Match? ☐ Yes ☐ No

5

Rewrite $2x + y = 6$ in slope-intercept form.

Answer: _____ Match? ☐ Yes ☐ No

6

Parallel to $y = 4x - 1$ and passes through $(0, 5)$

Answer: _____ Match? ☐ Yes ☐ No

7

Perpendicular to $y = -\frac{1}{2}x + 3$ and passes through $(0, -2)$

Answer: _____ Match? ☐ Yes ☐ No

8

A taxi charges a \$3 flat fee plus \$2 per mile. Write an equation for the cost y after x miles.

Answer: _____ Match? ☐ Yes ☐ No

Activity 3.2: Writing Equations of Lines

B

Name: _____ Partner's name: _____ Date: _____

Directions: Write the equation of each line in slope-intercept form, $y = mx + b$.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1

Slope 2, passes through $(0, -3)$

Answer: _____ Match? ☐ Yes ☐ No

2

Slope 3, passes through $(2, 8)$

Answer: _____ Match? ☐ Yes ☐ No

3

Passes through $(0, 4)$ and $(1, 7)$

Answer: _____ Match? ☐ Yes ☐ No

4

Passes through $(-1, -3)$ and $(2, 3)$

Answer: _____ Match? ☐ Yes ☐ No

5

Rewrite $4x + 2y = 12$ in slope-intercept form.

Answer: _____ Match? ☐ Yes ☐ No

6

Parallel to $y = 4x + 7$ and passes through $(1, 9)$

Answer: _____ Match? ☐ Yes ☐ No

7

Perpendicular to $y = -\frac{1}{2}x - 4$ and passes through $(1, 0)$

Answer: _____ Match? ☐ Yes ☐ No

8

A plant is 3 cm tall and grows 2 cm per week. Write an equation for its height y after x weeks.

Answer: _____ Match? ☐ Yes ☐ No